Power BI: Energy Consumption & Solar Production Analytics

1. Executive Summary

Role: Data Analyst / Business Intelligence Analyst

Tools: Microsoft Power BI, Power Query, DAX, Python (Seaborn), REST APIs

Data Sources:

- ESB (Electricity Supply Board) – Grid Import / Export Data
- SolisCloud – Solar Inverter & Battery System Data
- Meteostat API – Weather Data

This project delivers a complete end-to-end analytics solution designed to monitor and optimize residential energy consumption, solar generation, battery performance, and electricity costs.

By integrating multiple real-world data sources—including grid electricity usage, photovoltaic (PV) system performance, and weather conditions—the solution provides a unified analytical platform to track energy flows, evaluate system efficiency, and identify cost-saving opportunities.

The project demonstrates the full data lifecycle: **data ingestion, transformation, feature engineering, KPI development, visualization, and reporting**, all implemented within Power BI. The final dashboard enables continuous monitoring of energy usage patterns, solar contribution, and financial performance, supporting data-driven energy management decisions.

2. Business Problem & Objectives

Energy consumption costs and renewable energy adoption are increasingly influenced by usage behavior, weather patterns, and system performance. However, data from energy providers, solar systems, and weather services is typically stored in separate systems with inconsistent formats, making it difficult to understand overall energy efficiency and financial impact.

This project aimed to build a centralized analytics solution capable of:

- Tracking electricity import and export from the grid

- Monitoring solar generation and battery usage
- Calculating total electricity costs and savings
- Understanding energy consumption patterns by time
- Evaluating the impact of weather conditions on solar production
- Providing operational insights into energy efficiency

Key Objectives:

- Build a complete end-to-end data pipeline integrating multiple external data sources
- Transform and standardize raw energy data into a structured analytical model
- Develop advanced KPIs for energy usage, solar efficiency, and financial performance
- Deliver interactive dashboards for monitoring consumption, generation, and costs
- Enable time-based analysis for trend identification and forecasting readiness

3. The Process (Technical Implementation)

Step 1: Data Extraction & Transformation (ETL)

Data Sources Integration

The solution integrates three independent real-world data sources:

ESB (Electricity Supply Board)

- Grid electricity import and export data (kWh)
- Electricity tariff information
- Cost and revenue calculations

SolisCloud (Solar System Data)

- Solar generation data

- Battery charge and discharge activity
- Load consumption metrics
- System performance indicators

Meteostat API (Weather Data)

- Temperature measurements
- Sunshine duration
- Precipitation levels

Data Transformation (Power Query)

Significant transformation work was required to standardize the datasets.

Key transformations included:

- Cleaning missing and inconsistent records
- Standardizing timestamps and time zones
- Creating calculated time dimensions
- Removing duplicates and correcting data types
- Unpivoting SolisCloud data where energy flows were stored as separate columns
- Merging datasets from multiple sources into a unified structure
- Creating derived fields for analysis (day/night tariff classification, weather metrics)

This stage ensured reliable, analysis-ready datasets.

Step 2: Data Modeling

Designed a structured **Star Schema** data model consisting of **7 tables**.

Fact Tables

- ESB Energy Data
- Solar System Data (SolisCloud)
- Weather Data

Dimension Tables

- Energy Flow
- Time
- Calendar
- Measures Table

Key Modeling Features:

- One-to-Many relationships between fact and dimension tables
- Optimized model for performance and scalability
- Standardized time-based relationships for accurate time intelligence
- Clean semantic layer for reporting and analysis
- Hidden technical columns for improved usability

The model supports reliable calculations, efficient filtering, and scalable reporting.

Step 3: Advanced DAX Calculations

A comprehensive KPI framework was developed to monitor energy performance, financial impact, and system efficiency.

Energy Metrics

- ESB Grid Import

- ESB Grid Export
- Net Grid Energy
- Peak Grid Demand (kW)
- Average Grid Demand (kW)
- Minimum Grid Demand

Financial Metrics

- ESB Total Import Cost
- ESB Export Revenue
- Net Electricity Cost
- Cost per kWh
- Solar Savings
- Total Solar Value
- Solar Bill Coverage %

Solar & Battery Performance

- SOLIS PV Generation
- Solar Self Consumption
- Solar Self Consumption Rate
- Battery Throughput
- Battery Efficiency
- Battery Cycles
- Solar Contribution

- Grid Contribution
- Full Load Hours
- Capacity Factor

Time Intelligence Metrics

- Last Month Import Cost
- Last Year Import Cost
- MoM Import Cost Change
- YoY Import Cost Change
- YTD Import Cost
- Rolling 12-Month Import Cost Trend

Weather Metrics

- Average Temperature
- Sunshine Hours
- Total Precipitation
- PV per Sunshine Hour
- Average Daily Sunshine
- Rainy Days

These calculations enable detailed operational monitoring and financial analysis.

Step 4: Dashboard Design & Visualization

The final solution consists of **9 interactive report pages**, designed to provide both operational monitoring and strategic insights.

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1. Energy Overview (Executive Dashboard)

Provides a high-level summary of system performance:

- Grid Import and Export
- PV Solar Generation
- Battery Throughput
- Net Electricity Cost
- Energy Flow Trends
- Energy Source Distribution
- Electricity Import Patterns

Includes advanced analytics such as:

- Python Heatmap: Grid Import by Day vs Hour
- Weather Impact Visualization: Sunshine vs Solar Generation

2. Energy Overview – Detailed Metrics View

Provides focused analysis for individual energy metrics:

- Grid Import performance
- Day vs Night consumption
- Average Daily and Monthly usage
- Export performance trends
- PV generation monitoring

3. Solar & Battery Performance

Monitors renewable energy production and storage behavior:

- Solar generation vs load consumption
- Battery charge and discharge patterns
- Battery efficiency monitoring
- Solar savings calculation
- Total solar value estimation

4. Grid Import vs Export Analysis

Evaluates electricity usage and grid interaction:

- Import vs export comparison
- Day tariff breakdown
- Hourly and weekday consumption patterns
- Grid dependency monitoring

5. Cost & Revenue Analysis

Tracks financial performance:

- Total electricity cost
- Export revenue
- Net electricity cost
- Cumulative cost vs revenue trends
- Solar savings and value

6. Table Summary

Comprehensive dataset view including:

- All calculated metrics

- Time intelligence columns
- Aggregated performance indicators

Useful for validation and detailed analysis.

7. Daily Details (Drillthrough Page)

Provides detailed hourly analysis for specific dates:

- Import energy usage
- Export activity
- Solar generation
- Battery performance

Supports root cause investigation.

8. Time Intelligence Dashboard

Analyzes performance trends over time:

- YTD Import Cost
- Rolling 12-Month Import Cost Trend
- Month-over-Month Cost Changes
- Year-over-Year Cost Changes

Supports long-term planning and forecasting.

9. Weather Impact Analysis

Explores relationships between weather and energy performance:

- Temperature vs energy consumption
- Sunshine vs solar generation

- Precipitation vs load behavior
- Seasonal weather patterns

Supports predictive and operational planning.

4. Advanced Dashboard Design & User Experience (UX)

This project prioritizes usability, interactivity, and analytical depth.

Custom Navigation

- Implemented custom navigation bars using buttons and bookmarks
- Created an application-style user experience
- Applied standardized electricity-themed color palette

Interactive Features

- Drill Up and Drill Down navigation
- Drillthrough analysis pages
- Custom tooltips for contextual insights
- Managed visual interactions between charts
- Dynamic filtering and time-based slicing

Advanced Visualizations (Python Integration)

To extend analytical capability beyond standard visuals:

- Heatmap showing electricity import patterns by day and hour
- Advanced statistical visualizations using Python Seaborn

This demonstrates the ability to integrate Power BI with external analytical tools.

5. Key Insights Discovered

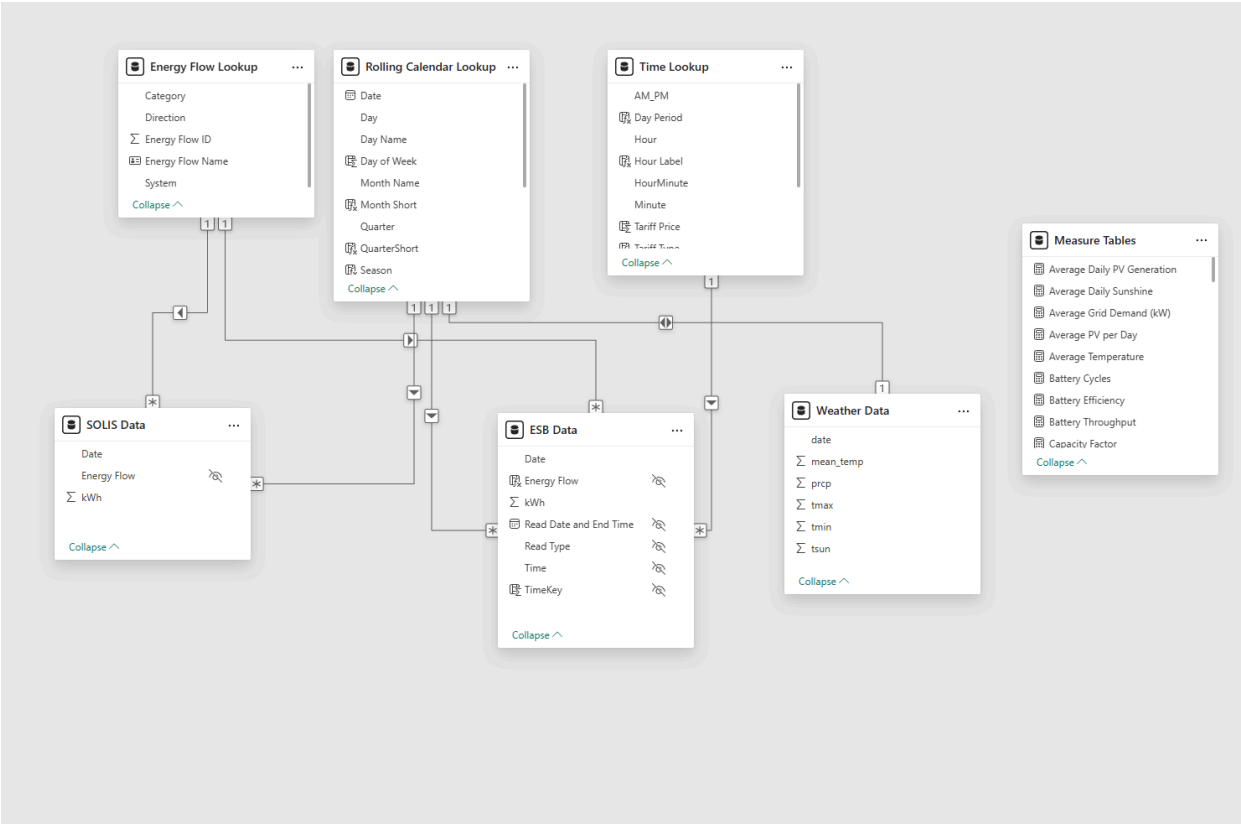
(*) Detailed analytical findings have been intentionally omitted to comply with data privacy and responsible data handling principles, as the dataset contains personally identifiable energy usage and financial information.

6. Visuals

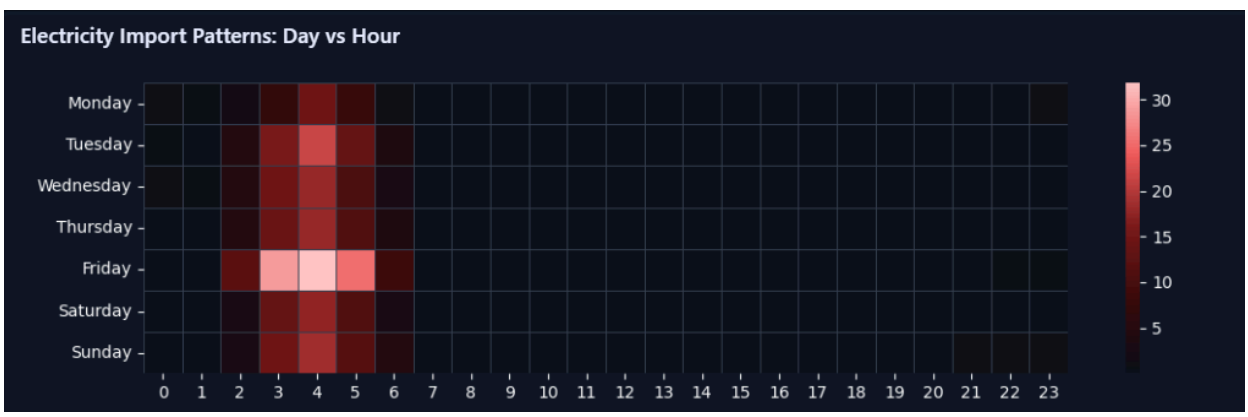
Recommended screenshots:

- Grid Import and Export

- Data Model (Star Schema)



- Python Heatmap Visualization



(*) Rest of images not included due reports contains personally identifiable energy usage